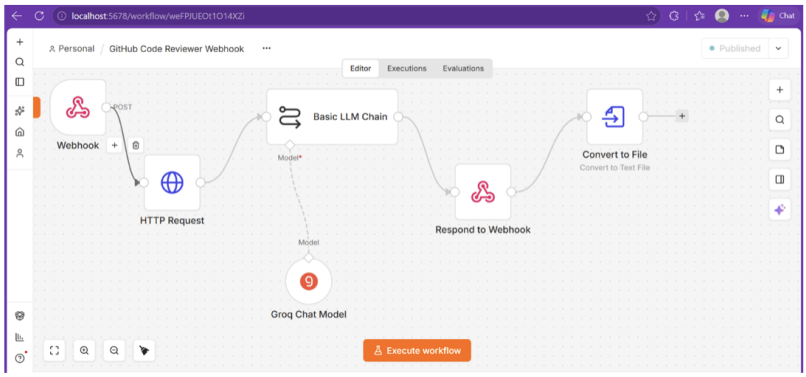
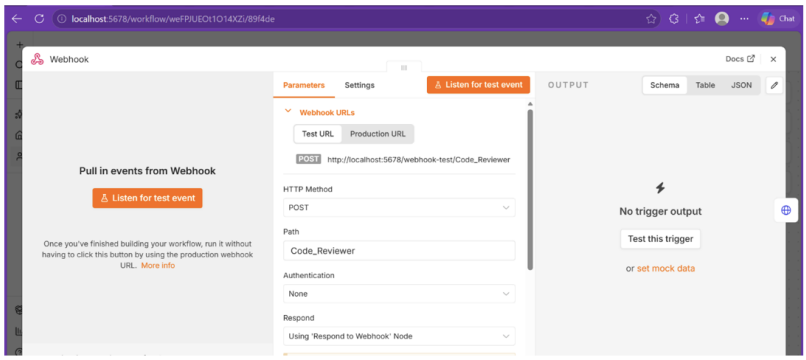
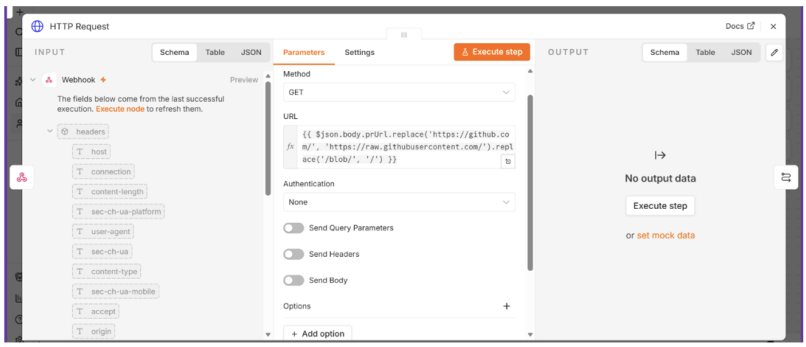
Week-6 [Day-2]

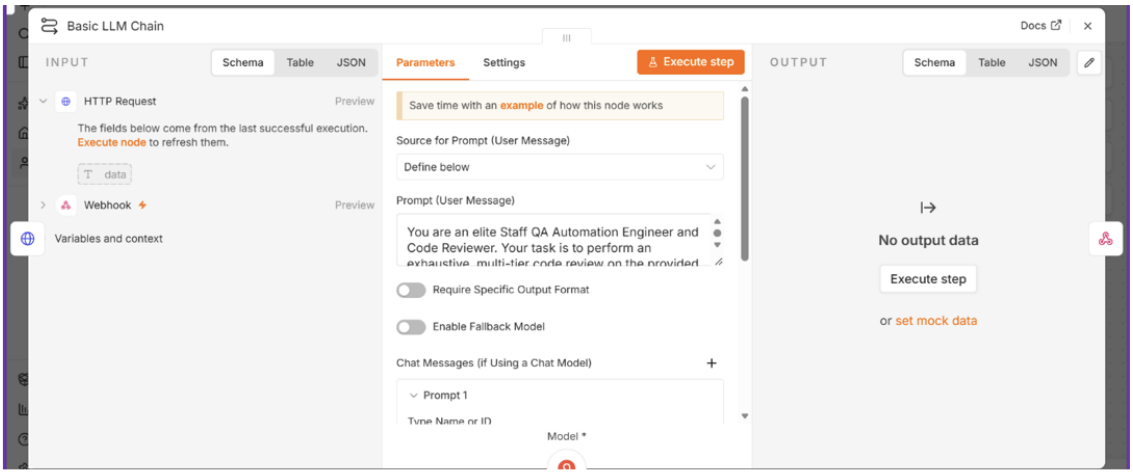
Home Assignment

GitHub Code Reviewer:









Prompt (User Message) in Basic LLM Chain:

You are an elite Staff QA Automation Engineer and Code Reviewer. Your task is to perform an exhaustive, multi-tier code review on the provided Selenium Java automation script.

Analyze the code across three critical vectors:

1. Security (Credential management, secure coding standards)

2. Performance & Reliability (Resource leaks, driver teardown, explicit/implicit wait optimization)

3. Code Quality (Naming conventions, structure, error handling robustness)

Output your evaluation strictly in a structured format so the downstream n8n nodes can parse it into JSON. You must provide:

- A numeric Security Score (0 to 100)

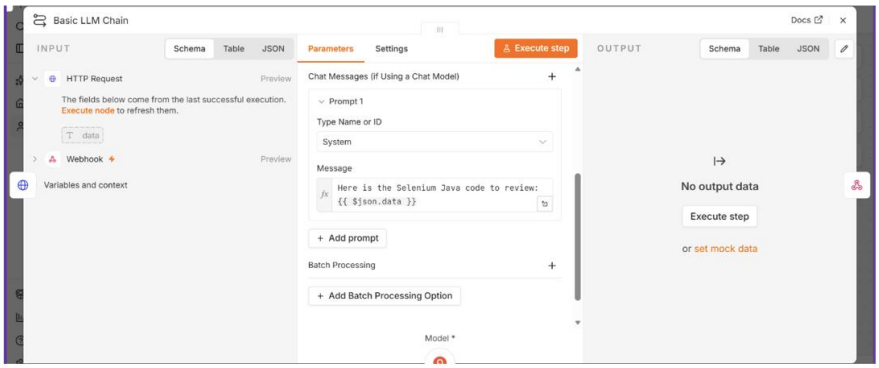
- A Performance Rating letter grade (A, B, C, D, or F)

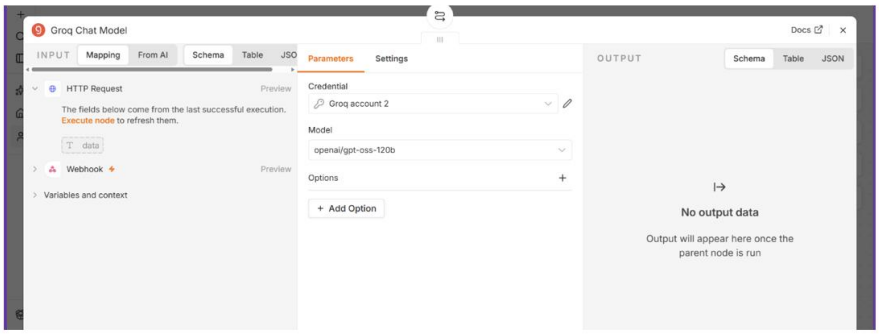
- A Code Quality classification status (Excellent, Good, Needs Improvement, or Critical)

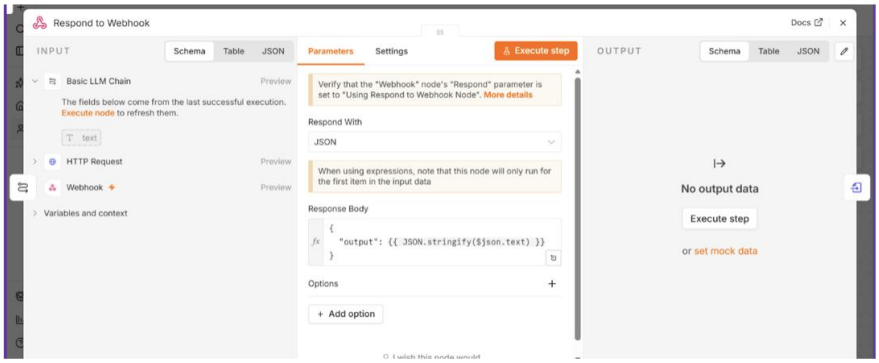
- A brief bulleted summary of findings

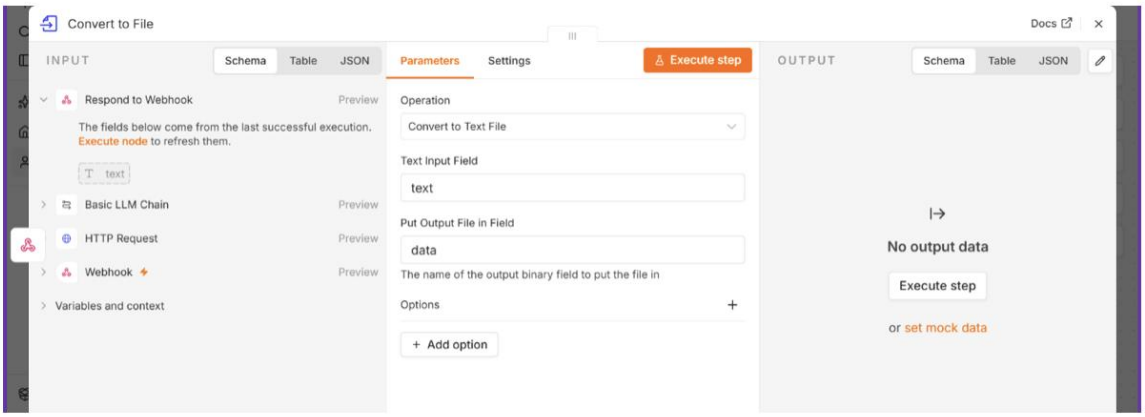
- An in-depth, line-by-line detailed feedback block formatted in clean Markdown snippets.

Ensure you explicitly look for dangling browser sessions, hardcoded credentials, and missing try-catch-finally isolation logic.









Prompt for Custom UI:

Act as an expert frontend engineer. Build a modern, clean, single-page React/Tailwind CSS dashboard for a GitHub Code Reviewer Agent.

Follow this phase-wise implementation plan:

Phase 1: Component Structure & UI State

- Create a layout with a dark-mode tech aesthetic (GitHub-inspired slate/gray tones) with Organization Name as Davidson & Co and designed a Logo based on Organization Name on Left side.

- Add an input form for a GitHub Repository URL or Pull Request URL, a dropdown to choose the programming language, and a "Run Review" button in Responsive Preview section.

- Create loading states (skeleton animations or progress tickers) and error states.

- Should be responsive for Desktop, Tablet & Mobile Devices (Both Android and iOS).

- Should have both Header and Footer.

Phase 2: Review Display Dashboard

- Design a split-screen or modular layout for the output.

- Left panel: Summary metrics (Security score, Performance rating, Code quality rating using visual badges).

- Right panel: Detailed feedback section that supports Markdown rendering for code snippets.

Phase 3: n8n Webhook Integration

- Add an asynchronous JavaScript fetch function that triggers on form submission.

- It must POST the form data as a JSON payload: { prUrl: value }

- The target endpoint should be the live n8n production URL: "http://localhost:5678/webhook/Code_Reviewer" (Note: match casing 'Code_Reviewer' exactly and do not use 'webhook-test' for live form submissions).

- Set headers to include 'Content-Type': 'application/json'.

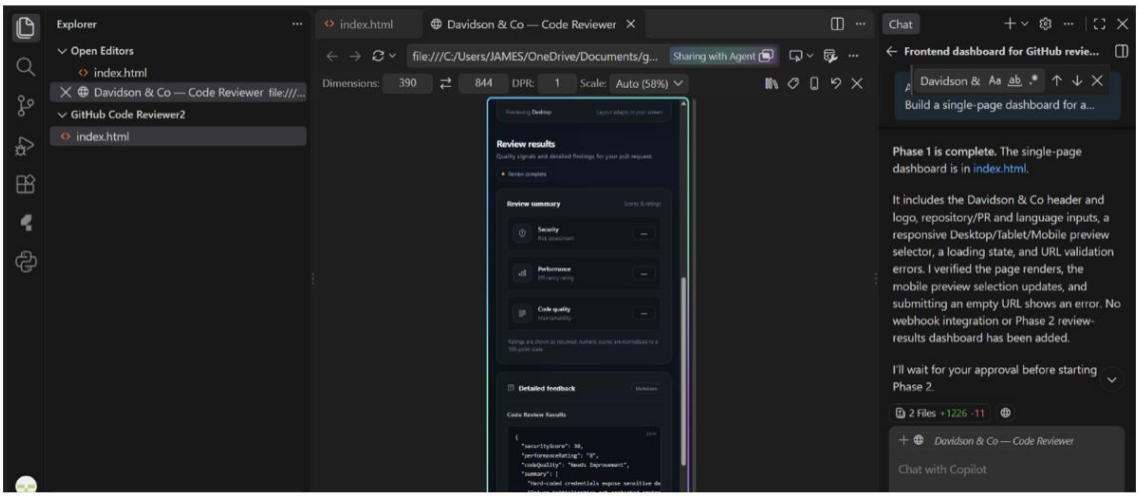
- Handle the incoming JSON response returned by the "Respond to Webhook" node and map the output string directly to the UI elements created in Phase 1 & 2.

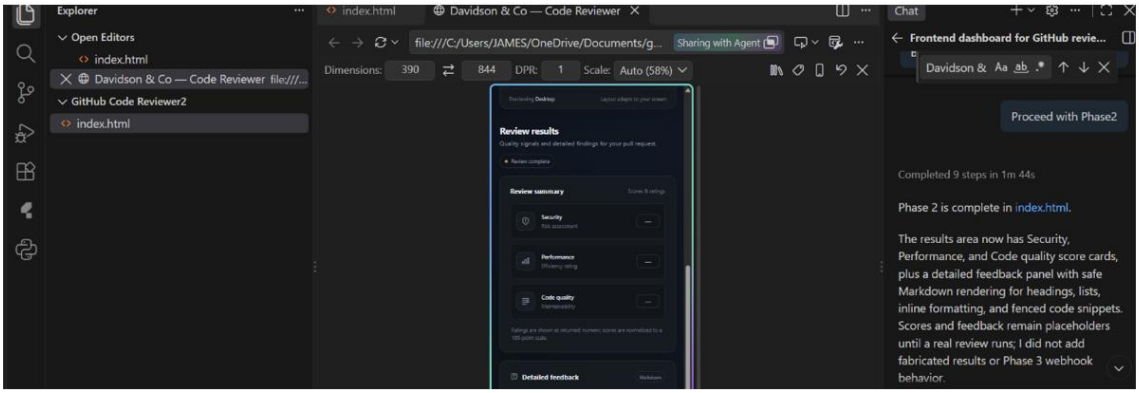
If the Search is blank, It should display proper error message. If the Search is not matches, then also should display error message.

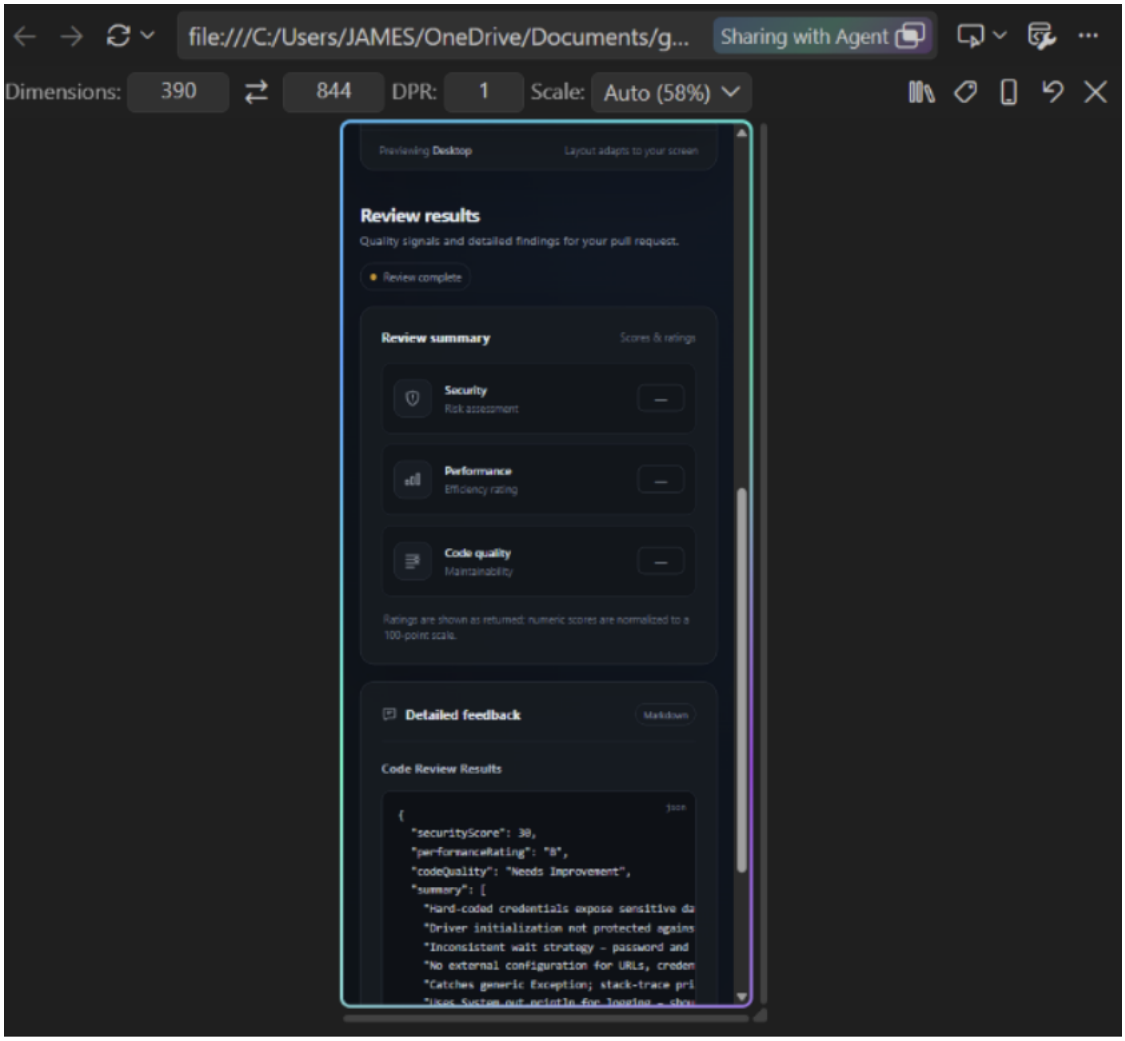
If the Search matches, then should analyze the code and provide the exact score.

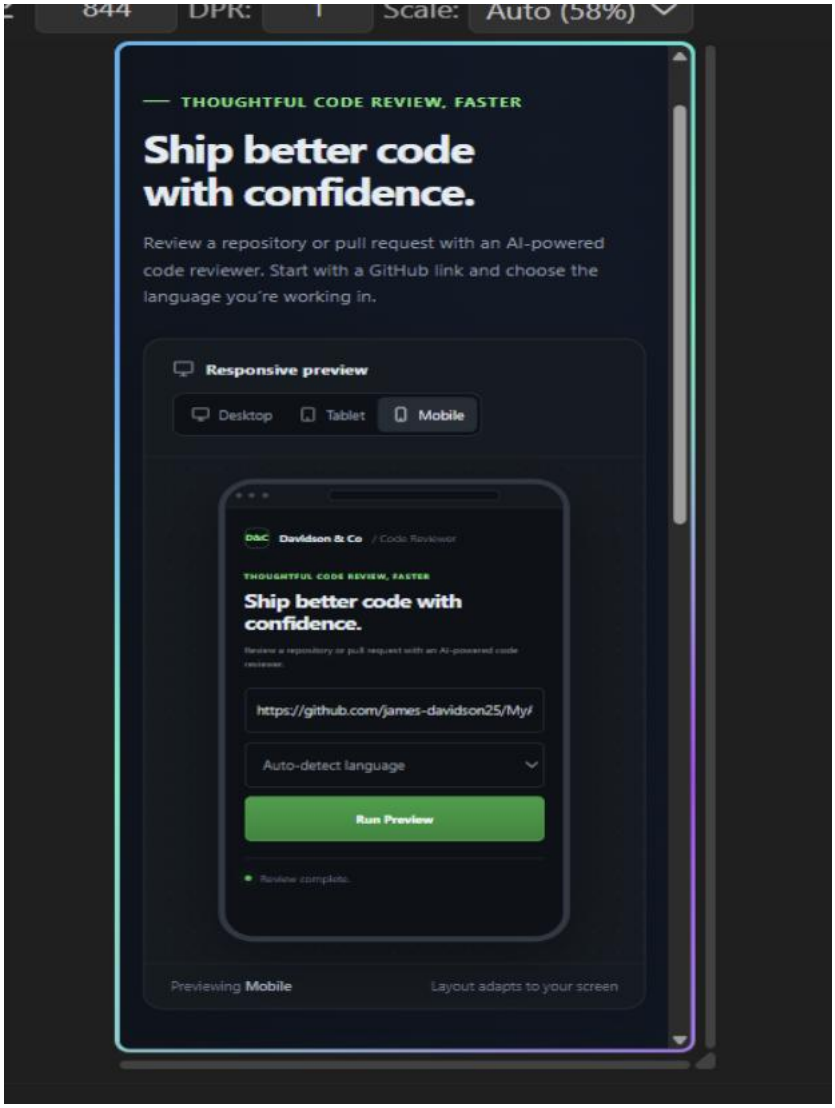
Strict phase rules Complete only one phase at a time. After Phase 1, stop and wait for your approval. Begin Phase 2 only after you explicitly approve Phase 1. Do not start the next phase without approval. Do not assume or generate unsupported requirements.

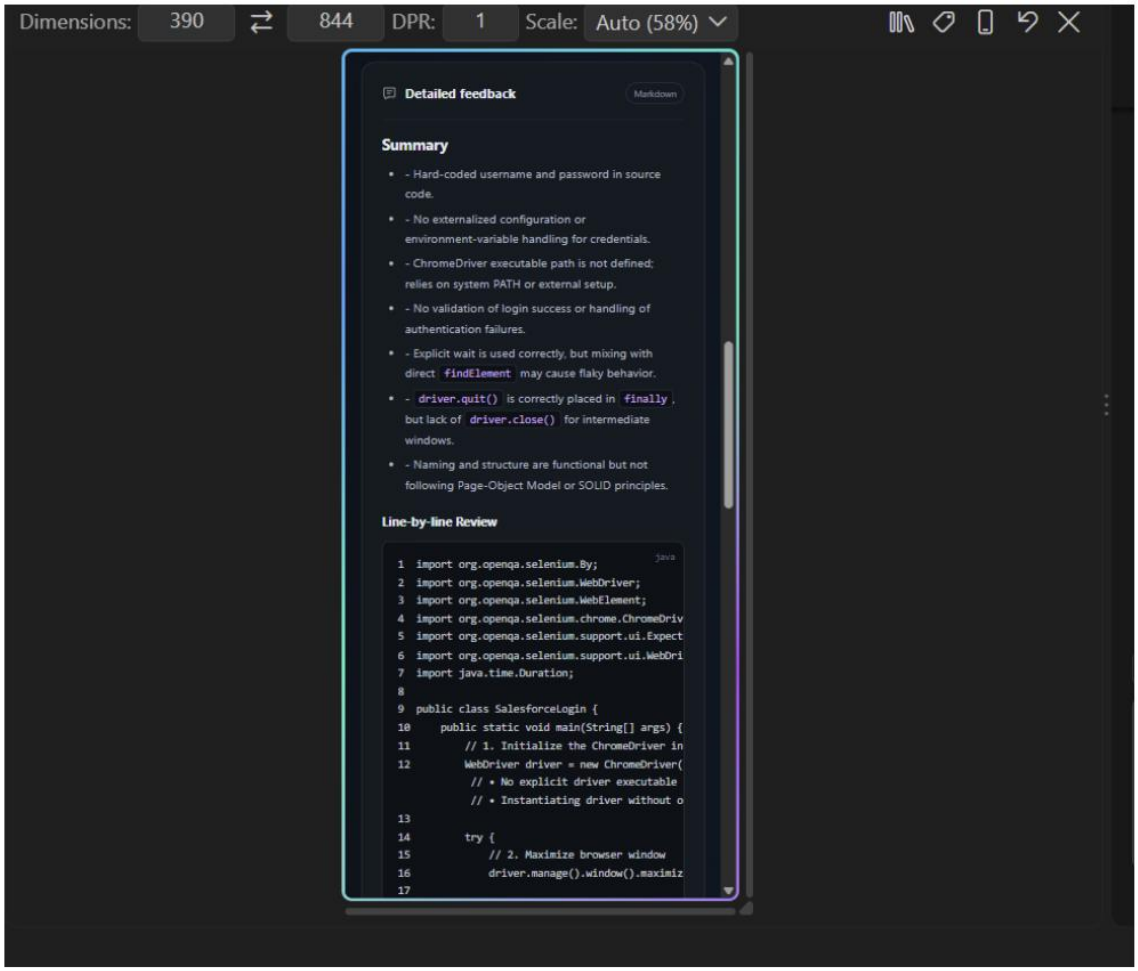
Phase-wise Implementation:

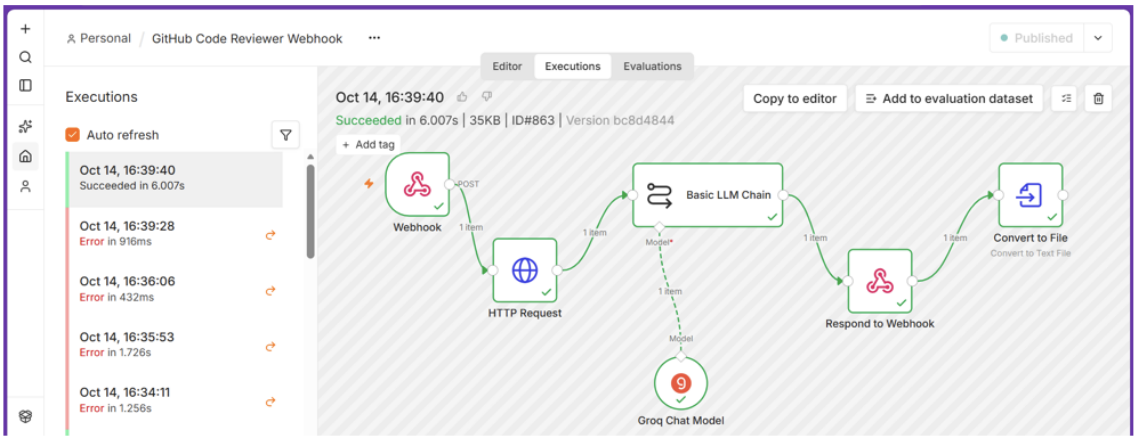

Explorer
Open Editors
index.html
Davidson & Co — Code Reviewer file:///...
GitHub Code Reviewer2
index.html
index.html
Davidson & Co — Code Reviewer
file:///C:/Users/JAMES/OneDrive/Documents/g...
Sharing with Agent
Dimensions: 390 844 DPR: 1 Scale: Auto (58%)
Review results
Review summary
Security
Performance
Code quality
Detailed feedback
Code Review Results
Chat
Frontend dashboard for GitHub revie...
Davidson &
Build a single-page dashboard for a...
Phase 1 is complete. The single-page dashboard is in index.html.
It includes the Davidson & Co header and logo, repository/PR and language inputs, a responsive Desktop/Tablet/Mobile preview selector, a loading state, and URL validation errors. I verified the page renders, the mobile preview selection updates, and submitting an empty URL shows an error. No webhook integration or Phase 2 review-results dashboard has been added.
I'll wait for your approval before starting Phase 2.
2 Files +1226 -11
Davidson & Co — Code Reviewer
Chat with Copilot

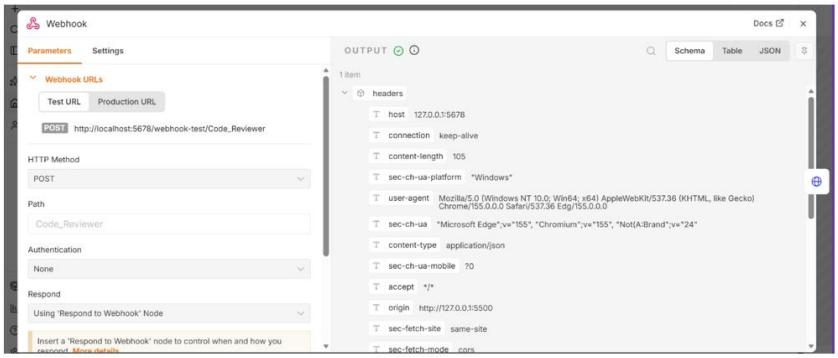

Explorer
Open Editors
index.html
Davidson & Co — Code Reviewer file:///...
GitHub Code Reviewer2
index.html
index.html
Davidson & Co — Code Reviewer
file:///C:/Users/JAMES/OneDrive/Documents/g...
Sharing with Agent
Dimensions: 390 844 DPR: 1 Scale: Auto (58%)
Review results
Review summary
Security
Performance
Code quality
Detailed feedback
Chat
Frontend dashboard for GitHub revie...
Davidson &
Proceed with Phase2
Completed 9 steps in 1m 44s
Phase 2 is complete in index.html.
The results area now has Security, Performance, and Code quality score cards, plus a detailed feedback panel with safe Markdown rendering for headings, lists, inline formatting, and fenced code snippets. Scores and feedback remain placeholders until a real review runs; I did not add fabricated results or Phase 3 webhook behavior.

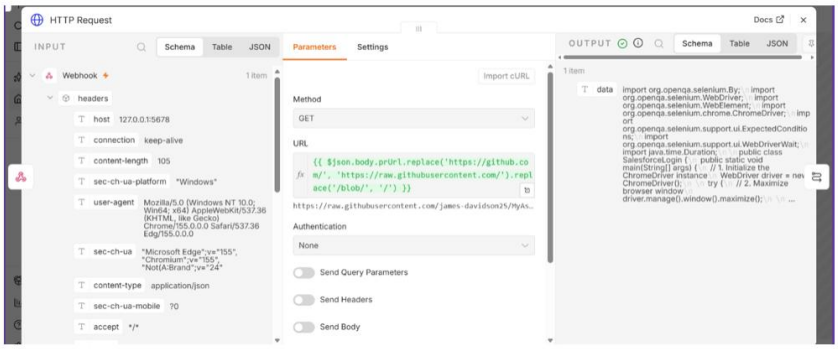

file:///C:/Users/JAMES/OneDrive/Documents/g...
Sharing with Agent
Dimensions: 390 844 DPR: 1 Scale: Auto (58%)
Previewing Desktop
Layout adapts to your screen
Review results
Quality signals and detailed findings for your pull request.
Review complete
Review summary
Scores & ratings
Security
Risk assessment
Performance
Efficiency rating
Code quality
Maintainability
Ratings are shown as returned; numeric scores are normalized to a 100-point scale.
Detailed feedback
Markdown
Code Review Results
json
{
"securityScore": 30,
"performanceRating": "B",
"codeQuality": "Needs Improvement",
"summary": [
"Hard-coded credentials expose sensitive da
"Driver initialization not protected agains
"Inconsistent wait strategy – password and
"No external configuration for URLs, creden
"Catches generic Exception; stack-trace pri

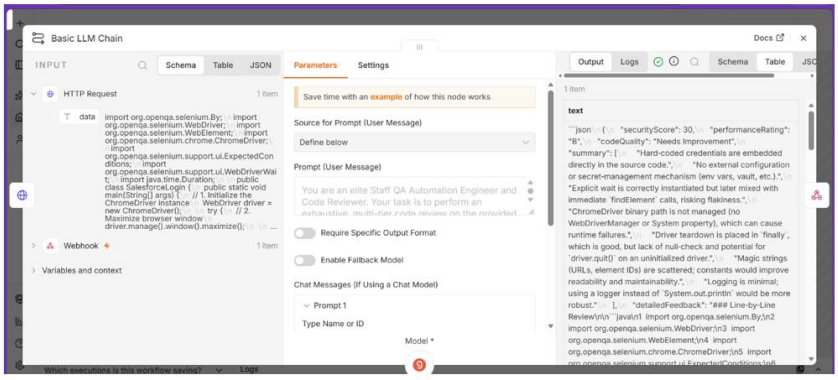

844
DPR:
1
Scale:
Auto (58%)
— THOUGHTFUL CODE REVIEW, FASTER
Ship better code with confidence.
Review a repository or pull request with an AI-powered code reviewer. Start with a GitHub link and choose the language you're working in.
Responsive preview
Desktop
Tablet
Mobile
D&C
Davidson & Co
/ Code Reviewer
THOUGHTFUL CODE REVIEW, FASTER
Ship better code with confidence.
https://github.com/james-davidson25/My/
Auto-detect language
Run Preview
Review complete.
Previewing Mobile
Layout adapts to your screen

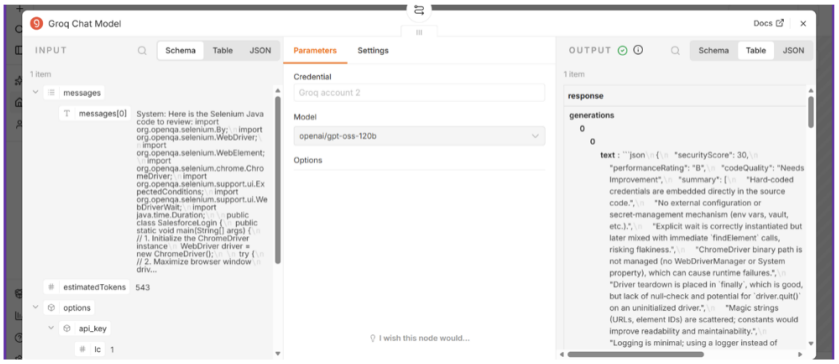

Dimensions: 390 844 DPR: 1 Scale: Auto (58%)
Detailed feedback
Markdown
Summary
- Hard-coded username and password in source code.
- No externalized configuration or environment-variable handling for credentials.
- ChromeDriver executable path is not defined; relies on system PATH or external setup.
- No validation of login success or handling of authentication failures.
- Explicit wait is used correctly, but mixing with direct findElement may cause flaky behavior.
- driver.quit() is correctly placed in finally, but lack of driver.close() for intermediate windows.
- Naming and structure are functional but not following Page-Object Model or SOLID principles.
Line-by-line Review
java
1 import org.openqa.selenium.By;
2 import org.openqa.selenium.WebDriver;
3 import org.openqa.selenium.WebElement;
4 import org.openqa.selenium.chrome.ChromeDriv
5 import org.openqa.selenium.support.ui.Expect
6 import org.openqa.selenium.support.ui.WebDri
7 import java.time.Duration;
8
9 public class SalesforceLogin {
10 public static void main(String[] args) {
11 // 1. Initialize the ChromeDriver in
12 WebDriver driver = new ChromeDriver(
// • No explicit driver executable
// • Instantiating driver without o
13
14 try {
15 // 2. Maximize browser window
16 driver.manage().window().maximiz
17

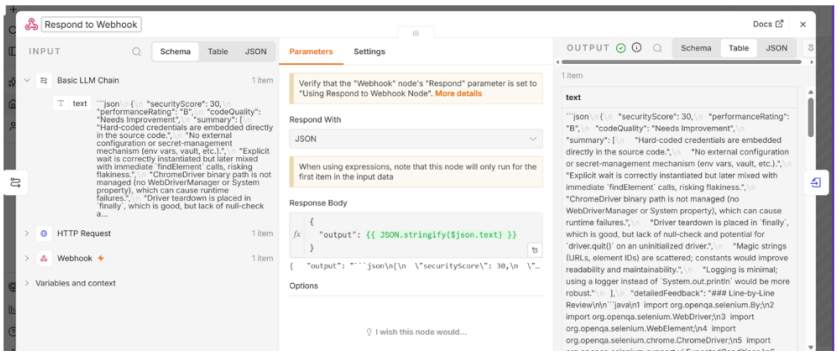

Personal / GitHub Code Reviewer Webhook
Published
Editor
Executions
Evaluations
Executions
Auto refresh
Oct 14, 16:39:40
Succeeded in 6.007s
Oct 14, 16:39:28
Error in 916ms
Oct 14, 16:36:06
Error in 432ms
Oct 14, 16:35:53
Error in 1.726s
Oct 14, 16:34:11
Error in 1.256s
Oct 14, 16:39:40
Copy to editor
Add to evaluation dataset
Succeeded in 6.007s | 35KB | ID#863 | Version bc8d4844
+ Add tag
Webhook
POST
1 item
HTTP Request
Basic LLM Chain
Model*
Model
Groq Chat Model
Respond to Webhook
Convert to File
Convert to Text File

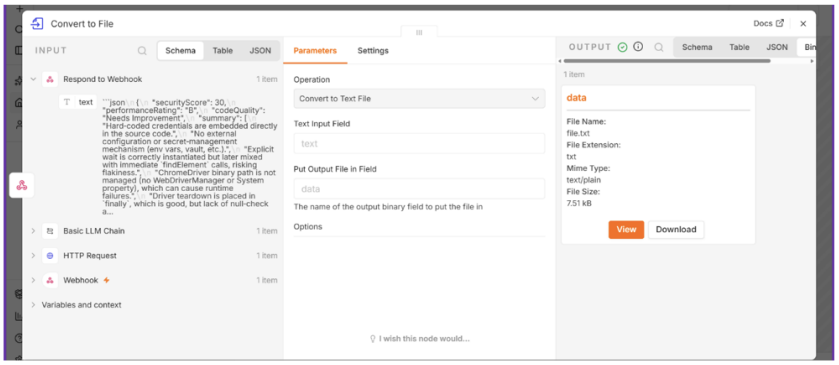

Webhook
Docs
Parameters
Settings
OUTPUT
Schema
Table
JSON
Webhook URLs
Test URL
Production URL
POST http://localhost:5678/webhook-test/Code_Reviewer
HTTP Method
POST
Path
Code_Reviewer
Authentication
None
Respond
Using 'Respond to Webhook' Node
Insert a 'Respond to Webhook' node to control when and how you
1 item
headers
host 127.0.0.1:5678
connection keep-alive
content-length 105
sec-ch-ua-platform "Windows"
user-agent Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/155.0.0.0 Safari/537.36 Edg/155.0.0.0
sec-ch-ua "Microsoft Edge";v="155", "Chromium";v="155", "Not(A:Brand";v="24"
content-type application/json
sec-ch-ua-mobile ?0
accept */*
origin http://127.0.0.1:5500
sec-fetch-site same-site
sec-fetch-mode cors

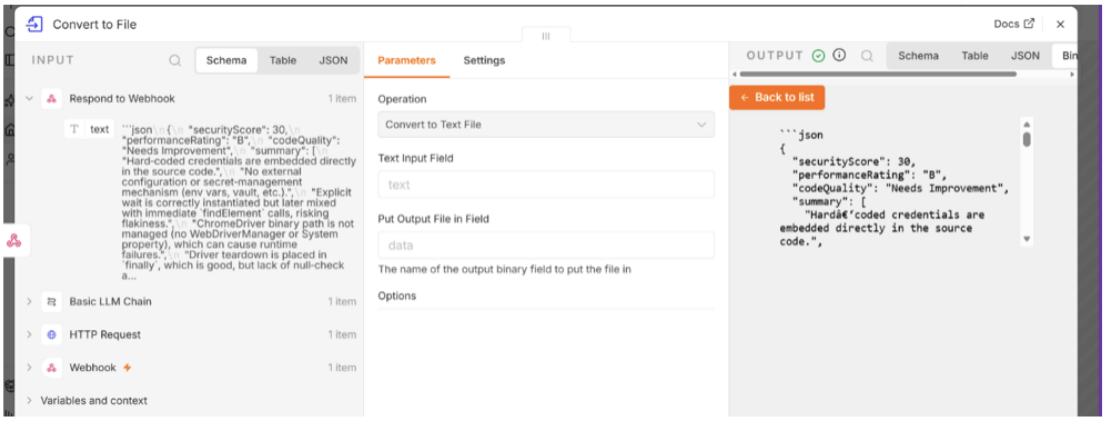

HTTP Request
Docs
INPUT
Schema
Table
JSON
Parameters
Settings
OUTPUT
Webhook
1 item
headers
host 127.0.0.1:5678
connection keep-alive
content-length 105
sec-ch-ua-platform "Windows"
user-agent Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/155.0.0.0 Safari/537.36 Edg/155.0.0.0
sec-ch-ua "Microsoft Edge";v="155", "Chromium";v="155", "Not(A:Brand";v="24"
content-type application/json
sec-ch-ua-mobile ?0
accept */*
Import cURL
Method
GET
URL
{{ $json.body.prUrl.replace('https://github.com/', 'https://raw.githubusercontent.com/').replace('/blob/', '/') }}
https://raw.githubusercontent.com/james-davidson25/MyAs...
Authentication
None
Send Query Parameters
Send Headers
Send Body
data import org.openqa.selenium.By; import org.openqa.selenium.WebDriver; import org.openqa.selenium.WebElement; import org.openqa.selenium.chrome.ChromeDriver; import org.openqa.selenium.support.ui.ExpectedConditions; import org.openqa.selenium.support.ui.WebDriverWait; import java.time.Duration; public class SalesforceLogin { public static void main(String[] args) { // 1. Initialize the ChromeDriver instance WebDriver driver = new ChromeDriver(); try { // 2. Maximize browser window driver.manage().window().maximize();


Basic LLM Chain
Docs
INPUT
Schema
Table
JSON
Parameters
Settings
Output
Logs
HTTP Request
1 item
data import org.openqa.selenium.By; import org.openqa.selenium.WebDriver; import org.openqa.selenium.WebElement; import org.openqa.selenium.chrome.ChromeDriver; import org.openqa.selenium.support.ui.ExpectedConditions; import org.openqa.selenium.support.ui.WebDriverWait; import java.time.Duration; public class SalesforceLogin { public static void main(String[] args) { // 1. Initialize the ChromeDriver instance WebDriver driver = new ChromeDriver(); try { // 2. Maximize browser window driver.manage().window().maximize();
Webhook
1 item
Variables and context
Save time with an example of how this node works
Source for Prompt (User Message)
Define below
Prompt (User Message)
You are an elite Staff QA Automation Engineer and Code Reviewer. Your task is to perform an
Require Specific Output Format
Enable Fallback Model
Chat Messages (If Using a Chat Model)
Prompt 1
Type Name or ID
Model *
Which executions is this workflow saving?
Logs
1 item
text
```json { "securityScore": 30, "performanceRating": "B", "codeQuality": "Needs Improvement", "summary": ["Hard-coded credentials are embedded directly in the source code.", "No external configuration or secret-management mechanism (env vars, vault, etc.).", "Explicit wait is correctly instantiated but later mixed with immediate `findElement` calls, risking flakiness.", "ChromeDriver binary path is not managed (no WebDriverManager or System property), which can cause runtime failures.", "Driver teardown is placed in `finally`, which is good, but lack of null-check and potential for `driver.quit()` on an uninitialized driver.", "Magic strings (URLs, element IDs) are scattered; constants would improve readability and maintainability.", "Logging is minimal; using a logger instead of `System.out.println` would be more robust."], "detailedFeedback": "### Line-by-Line Review\n\n```java\n1 import org.openqa.selenium.By;\n2 import org.openqa.selenium.WebDriver;\n3 import org.openqa.selenium.WebElement;\n4 import org.openqa.selenium.chrome.ChromeDriver;\n5 import
```


Groq Chat Model
Docs
INPUT
Schema
Table
JSON
Parameters
Settings
Credential
Groq account 2
Model
openai/gpt-oss-120b
Options
OUTPUT
response
generations
messages
estimatedTokens
543
options
api_key


Respond to Webhook
Docs
INPUT
Schema
Table
JSON
Parameters
Settings
Basic LLM Chain
HTTP Request
Webhook
Variables and context
Verify that the "Webhook" node's "Respond" parameter is set to "Using Respond to Webhook Node". More details
Respond With
JSON
When using expressions, note that this node will only run for the first item in the input data
Response Body
{ "output": {{ JSON.stringify($json.text) }} }
Options
OUTPUT
text


Convert to File
Docs
INPUT
Schema
Table
JSON
Parameters
Settings
Respond to Webhook
Basic LLM Chain
HTTP Request
Webhook
Variables and context
Operation
Convert to Text File
Text Input Field
text
Put Output File in Field
data
The name of the output binary field to put the file in
Options
OUTPUT
data
File Name:
file.txt
File Extension:
txt
Mime Type:
text/plain
File Size:
7.51 kB
View
Download


Convert to File
Docs
INPUT
Schema
Table
JSON
Parameters
Settings
Respond to Webhook
Basic LLM Chain
HTTP Request
Webhook
Variables and context
Operation
Convert to Text File
Text Input Field
text
Put Output File in Field
data
The name of the output binary field to put the file in
Options
OUTPUT
Back to list
```json
{
"securityScore": 30,
"performanceRating": "B",
"codeQuality": "Needs Improvement",
"summary": [
"Hard-coded credentials are embedded directly in the source code.",
```